

Engineering Document E06 — HQ UI Component Library & Design System

Product Name: HQ

Engineering Package: Version 1.0

Purpose

This document defines the complete UI Component Library and Design System for HQ.

The objective is to create a consistent, accessible, reusable, and premium user interface that reflects HQ's identity as an AI Executive Headquarters.

Every screen in HQ should feel like part of one cohesive product rather than a collection of individual pages.

Design Principles

HQ should communicate:

- Intelligence
- Professionalism
- Precision
- Trust
- Speed
- Simplicity
- Premium quality

Every component must support these principles.

Technology Stack

UI Foundation

- shadcn/ui

Styling

- Tailwind CSS

Icons

- Lucide React

Animation

- Framer Motion

Charts

- Recharts

Fonts

- Inter (UI)
 - JetBrains Mono (technical content)
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Component Architecture

Foundation Components

↓

Layout Components

↓

Business Components

↓

AI Components

↓

Dashboard Components

↓

Page Templates

Each layer builds upon the previous one.

Foundation Components

These are the smallest reusable building blocks.

Buttons

Variants:

- Primary
- Secondary
- Outline
- Ghost
- Link
- Destructive
- Success

States:

- Default
 - Hover
 - Focus
 - Active
 - Loading
 - Disabled
-

Inputs

Support:

- Text

- Email
- Password
- Number
- Search
- URL
- Multiline

Validation:

- Success
- Error
- Warning

Form Components

Include:

- Checkbox
- Radio
- Switch
- Select
- Multi Select
- Date Picker
- Time Picker
- File Upload

Cards

Types:

- Standard
 - Elevated
 - Interactive
 - Analytics
 - Executive
 - Mission
-

Badges

Variants:

- Success
 - Warning
 - Error
 - Info
 - Neutral
 - AI
 - Premium
-

Avatars

Support:

- User
- Executive
- Organization

Fallback initials are displayed automatically.

Dialog Components

Include:

- Modal
 - Drawer
 - Confirmation Dialog
 - Alert Dialog
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Layout Components

These define page structure.

Include:

- App Shell
 - Sidebar
 - Navigation Bar
 - Top Header
 - Footer
 - Dashboard Grid
 - Page Container
 - Section Header
-

Navigation Components

Support:

- Sidebar Navigation
 - Breadcrumbs
 - Tabs
 - Pagination
 - Navigation Rail
 - Command Palette
-

Dashboard Components

Widgets include:

- KPI Card
- Statistics Card
- Mission Summary
- Activity Feed
- Executive Status
- Revenue Chart
- Usage Chart
- Recent Assets

Each widget is independently reusable.

Executive Components

HQ introduces custom AI components.

Executive Card

Displays:

- Executive Avatar
 - Department
 - Status
 - Current Mission
 - Confidence
 - Activity
-

Executive Panel

Contains:

- Profile
 - Responsibilities
 - Active Tasks
 - Recommendations
 - Memory Summary
-

Executive Status Badge

States:

- Idle
 - Planning
 - Working
 - Reviewing
 - Waiting
 - Completed
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Mission Components

Mission UI includes:

Mission Card

Mission Timeline

Mission Progress

Mission Status

Mission Activity

Mission History

Mission Output Viewer

AI Components

These are unique to HQ.

Thinking Indicator

Animated visualization indicating AI reasoning is in progress.

Collaboration Timeline

Shows executives collaborating chronologically.

Executive Discussion Panel

Displays structured communication between executives.

AI Response Block

Displays:

- Executive
- Response
- Confidence
- Timestamp

Executive Approval Card

Shows:

Approved

Needs Revision

Rejected

Awaiting Review

Analytics Components

Charts include:

- Line Chart
- Bar Chart
- Pie Chart
- Area Chart
- Heatmap
- Activity Timeline

All charts share consistent spacing, typography, and interaction patterns.

Data Display Components

Include:

- Table
 - Data Grid
 - List
 - Timeline
 - Tree View
 - Accordion
 - Expandable Panels
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Feedback Components

Support:

- Toast Notifications
 - Progress Indicators
 - Skeleton Loaders
 - Empty States
 - Success Messages
 - Error Messages
 - Warning Messages
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Loading Components

Loading experiences include:

- Skeleton Cards
- Skeleton Dashboard
- Executive Loading State
- Mission Loading Timeline
- Full Page Loader

Loading should always indicate meaningful progress.

Empty States

Every empty screen should include:

- Illustration
- Description
- Call-to-action
- Helpful guidance

No page should appear unfinished.

Motion System

Animations communicate state.

Examples:

- Card Hover
- Sidebar Expand
- Executive Activation
- Mission Progress
- Notification Entrance
- Dashboard Transitions

Animations should be subtle and purposeful.

Accessibility

Every component supports:

- Keyboard navigation
- Screen readers
- Focus visibility
- ARIA labels
- High contrast
- Reduced motion

Accessibility is mandatory.

Responsive Behavior

Components adapt to:

- Mobile
- Tablet
- Laptop
- Desktop
- Ultra-wide displays

Layouts remain usable at all supported sizes.

Component States

Every interactive component defines:

- Default
- Hover
- Focus
- Active
- Disabled
- Loading
- Success
- Error

State behavior remains consistent across the application.

Component Documentation

Every component includes:

- Purpose
- Props

- Variants
- Accessibility guidance
- Usage examples
- Design notes

Documentation lives alongside implementation.

Component Testing

Each component requires:

- Unit tests
- Accessibility tests
- Visual regression tests
- Interaction tests

Reusable components must be reliable before adoption.

Future Expansion

The component library is designed to support:

- Mobile components
- Desktop application
- White-label themes
- Enterprise branding
- Additional AI visualizations

These additions should integrate without breaking existing components.

Success Criteria

The HQ UI Component Library succeeds when:

- The interface feels cohesive.

- Components are reusable.
 - Accessibility requirements are met.
 - Development is accelerated through reuse.
 - New features can be assembled without creating duplicate UI elements.
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Conclusion

The HQ UI Component Library and Design System establishes a unified visual and interaction language for the platform.

By combining reusable foundations, business-focused components, and AI-specific interfaces, HQ delivers a premium user experience that distinguishes it from generic dashboards while remaining scalable, maintainable, and ready for long-term product evolution.